# CSC144V Autonomous Vehicles
# Fall 2026 Midterm Exam ANS

Student Name:_____ ID：_____

| Total pts | |
|---|---|

Note: This exam covers L1.1 – L4.1 and the IoU/NMS part of L5. For Q2-Q6, **show your work.** Correct intermediate steps may earn partial credit even if the final answer is incorrect. For Q1, only the answer keys entered in the table count.

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 |
|---|---|---|---|---|---|
| /40 | /10 | /10 | /20 | /10 | /10 |

**Q1 (40 pts) Multiple-choice questions (2 pts each): enter your answer keys here:**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| B | C | A | B | B | A | D | D | C | C |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| A | A | B | D | C | D | B | A | C | D |

For the following multiple-choice questions, each question has exactly one correct answer key. Fill in the answer keys in the tables above. **(Answer keys written in the question area will not be counted.)**

1. Which statement correctly contrasts the two paths toward Level 4/5 described in the presentation?
A. Waymo begins with mass-deployed Level 2, while Tesla begins with Level 4 robotaxis.
B. Tesla starts with mass-deployed Level 2 and advances upward, while Waymo and nuTonomy start with Level 4 in a limited ODD and expand it.
C. Tesla begins with Level 5 in a limited ODD, while Waymo begins with Level 1.
D. Both approaches begin with Level 3 and avoid public deployment.

ANS: B. Tesla starts with mass-deployed Level 2 and advances upward, while Waymo and nuTonomy start with Level 4 in a limited ODD and expand it. [L1.1 Quiz Q8]

2. Why should total miles driven not be used as the sole safety metric?
A. Companies must drive the same routes and conditions.
B. Mileage cannot be recorded by autonomous vehicles.
C. Miles can involve very different driving conditions, and companies may avoid difficult conditions.
D. Only highway mileage is reported.

ANS: C. Miles can involve very different driving conditions, and companies may avoid difficult conditions. [L1.1 Quiz Q12]

3. What is the output of local planning?
A. A safe and smooth trajectory expressed as vehicle pose over time
B. A destination selected by the passenger
C. Throttle, brake, and steering actuator signals
D. A list of raw camera pixels

ANS: A. A safe and smooth trajectory expressed as vehicle pose over time [L1.1 Quiz Q15]

4. Which statement correctly describes semantic segmentation?
A. It assigns every pixel to a specific object instance such as car 1 or car 2.
B. It assigns each pixel to a semantic category such as road, car, or sky.
C. It assigns one label to an entire image.
D. It estimates only the distance to each object.

ANS: B. It assigns each pixel to a semantic category such as road, car, or sky. [L1.2 Quiz Q4]

5. What is a major limitation of IMU-based estimation?
A. It produces data only at 10 Hz.
B. It is prone to drift and accumulated errors over time.
C. It cannot operate in poor lighting.
D. It requires a lidar point cloud to measure acceleration.

ANS: B. It is prone to drift and accumulated errors over time. [L1.2 Quiz Q29]

6. How do HD maps differ from ordinary navigation maps such as Google Maps?
A. They provide a highly accurate, centimeter-level 3D representation designed for computer processing.
B. They are intended only for human viewing.
C. They are updated continuously by every vehicle.
D. They contain no road information.

ANS: A. They provide a highly accurate, centimeter-level 3D representation designed for computer processing. [L1.2 Quiz Q44]

7. What is a drawback of using a large GPU batch for inference?
A. It always lowers throughput.
B. It converts the GPU into an FPGA.
C. It prevents SIMD computation.
D. It can produce high and nondeterministic latency for an individual data item.

ANS: D. It can produce high and nondeterministic latency for an individual data item. [L1.3 Quiz Q10]

8. In an OTA update using a Connectivity ECU, when is the update loaded to the target ECU?
A. Only when all ECUs are powered off
B. Before the software is wirelessly transferred
C. Continuously while the vehicle is driving at full speed
D. When the vehicle is in a safe state

ANS: D. When the vehicle is in a safe state [L1.3 Quiz Q42]

9. According to the presentation, what caused recent AV accidents rather than trolley-problem-style ethical choices?
A. Moral Machine survey responses
B. Cultural preferences programmed into the vehicle
C. Failures in sensors or perception algorithms
D. Deliberate choices to sacrifice passengers

ANS: C. Failures in sensors or perception algorithms [L1.3 Quiz Q49]

10. Which statement correctly contrasts training and inference?
A. Training always runs on an edge device, whereas inference requires a GPU cluster.
B. Training and inference both update model weights in the same way.
C. Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass.
D. Training uses one forward pass, whereas inference requires millions of back-propagation iterations.

ANS: C. Training adjusts model parameters through repeated forward and backward passes, whereas inference typically uses one forward pass. [L2 Quiz Q5]

11. What does the SoftMax output represent in a multi-class classifier?
A. A list of class probabilities that sum to 1.0
B. A list containing only ground-truth labels
C. A list of hidden-layer sizes
D. A list of gradients whose values sum to zero

ANS: A. A list of class probabilities that sum to 1.0 [L2 Quiz Q14]

12. How does mini-batch stochastic gradient descent compute a parameter update?
A. It uses a small portion of the training data to compute the gradient.
B. It always uses the entire dataset for every gradient.
C. It uses validation data as ground-truth test data.
D. It updates parameters without calculating a loss.

ANS: A. It uses a small portion of the training data to compute the gradient. [L2 Quiz Q18]

13. In 5-fold cross-validation, how is each fold used?
A. One fold is permanently discarded.
B. Each fold serves as validation once while the other four serve as training data.
C. Each fold is used only for training and never for validation.
D. All five folds are used only for testing.

ANS: B. Each fold serves as validation once while the other four serve as training data. [L2 Quiz Q20]

14. For a convolution on an RGB input, what must equal the filter depth?
A. The output width
B. The number of filters
C. The filter width
D. The number of input channels

ANS: D. The number of input channels [L3 Quiz Q4]

15. What is the main benefit of dilated convolution described in the slides?
A. It removes the need for filters
B. It converts an image into a class label
C. It increases the receptive field without increasing the parameter count
D. It always reduces the number of channels to one

ANS: C. It increases the receptive field without increasing the parameter count [L3 Quiz Q10]

16. How does a CNN generally differ from an MLP in parameter usage?
A. An MLP uses convolution in every layer
B. They always have exactly the same number of parameters
C. A CNN always has more parameters
D. A CNN typically uses fewer parameters through local connectivity and sharing

ANS: D. A CNN typically uses fewer parameters through local connectivity and sharing [L3 Quiz Q15]

17. What is normally the first modification when applying transfer learning to a pretrained CNN for a new classification task?
A. Delete all convolution layers
B. Replace the SoftMax classification head with one for the new task
C. Replace every filter with a handcrafted edge detector
D. Increase the input to an arbitrary number of channels

ANS: B. Replace the SoftMax classification head with one for the new task [L3 Quiz Q20]

18. Compared with FGSM at the same $\varepsilon$, what does PGD with small steps typically achieve?
A. A higher loss inside the same $\varepsilon$-ball, paid for with several gradient evaluations, not one
B. A perturbation that leaves the $\varepsilon$-ball, since projection is applied only once at the very end
C. The same loss with a smaller perturbation, using one gradient evaluation like FGSM does here
D. Exactly the same $\delta$ as FGSM, because the projection at each step erases all of the differences

ANS: A. A higher loss inside the same $\varepsilon$-ball, paid for with several gradient evaluations, not one [L4.1 Quiz Q10]

19. An attacker owns the target model's training set but not its weights. Which strategy does the lecture describe?
A. Recover the target's weights by inverting its softmax outputs on the known training data set
B. Perturb the training set itself, so the deployed target loses accuracy without any transfer
C. Train a proxy white-box model, attack it, and rely on the attacks transferring to the target
D. Query the target with random inputs until its API returns the gradients of its own loss function

ANS: C. Train a proxy white-box model, attack it, and rely on the attacks transferring to the target [L4.1 Quiz Q16]

20. How does a Phantom attack on an ADAS differ from the ε-ball perturbations earlier in the lecture?
A. It adds $\ell 2$-bounded noise to every pixel of every camera frame so humans cannot see it at all
B. It shifts LiDAR returns by a bounded ε so that the range estimates move inside an $\ell\infty$ norm ball
C. It rewrites the ADAS model weights through a firmware update delivered by a roadside radio unit
D. A depthless picture shown for a few frames, with no $\ell p$ bound, placed where viewers do not look

ANS: D. A depthless picture shown for a few frames, with no $\ell p$ bound, placed where viewers do not look [L4.1 Quiz Q18]

**Q2. (10 pts) AV Overview and Sensors (L1.1 – L1.2)**

**(a) (6 pts) Sensor data rates.** A 128-channel lidar produces 1,800 azimuth measurements per channel per revolution and rotates at 10 Hz; each point takes 16 bytes. A camera has 1920 × 1080 pixels, 3 bytes per pixel (uncompressed RGB) and runs at 30 frames per second. Use 1 MB = $10^6$ bytes.

(i) (2 pts) Data rate of the lidar in MB/s.

(ii) (2 pts) Data rate of one camera in MB/s.

(iii) (2 pts) An AV carries 8 such cameras and 1 such lidar. What is the total data rate in GB/s, and how many TB of raw data are produced in one hour of driving?

**(b) (4 pts) Blind curve and V2X.** A disabled vehicle is stopped behind a blind curve. Your AV drives at 30 m/s (108 km/h). Its onboard sensors can first detect the stopped vehicle at a distance of 60 m. The total latency from detection to the start of braking is 0.2 s, and the AV then brakes with a constant deceleration of 6 m/s².

(i) (2 pts) Compute the total stopping distance.

(ii) (2 pts) Can the AV stop in time using onboard sensors only? Name one V2X communication that could help, and explain why in one sentence.

ANS:

(a) (i) 128 × 1,800 × 10 = 2,304,000 points/s; × 16 B = 36,864,000 B/s ≈ **36.9 MB/s** (≈ 295 Mbit/s). (1 pt points/s, 1 pt bytes/s)

(ii) 1920 × 1080 × 3 B = 6,220,800 B per frame; × 30 fps = 186,624,000 B/s ≈ **186.6 MB/s**. (1 pt bytes/frame, 1 pt per second)

(iii) 8 × 186.6 MB/s + 36.9 MB/s ≈ 1,529.9 MB/s ≈ **1.53 GB/s**; × 3,600 s ≈ 5.51 × $10^{12}$ B ≈ **5.5 TB per hour**. (1 pt total rate, 1 pt per hour) Note: one raw camera stream is about 5× the lidar stream, so cameras dominate the raw data volume.

(b) (i) Distance during the latency: 30 × 0.2 = 6 m. Braking distance: $v^2/(2a) = 30^2/(2 \times 6)$ = 900/12 = 75 m. Total stopping distance = 6 + 75 = **81 m**. (1 pt each term)

Formula: $d = v^2 / (2a)$, with v the speed at brake onset (m/s) and a the constant deceleration ($m/s^2$, > 0). From $v_f^2 = v_0^2 - 2ad$ with $v_f = 0$. Total stopping distance = $v \cdot t + v^2 / (2a)$, where t is the latency before braking starts; d grows with $v^2$ (2× speed → 4× distance).

(ii) **No**: 81 m > 60 m. After the 6 m reaction distance only $d_{remain} = d_{detect} - v \cdot t = 60 - 6$ = 54 m is left for braking, so the AV still hits the vehicle at $v_{impact} = \sqrt{(v_0^2 - 2 \cdot a \cdot d_{remain})} = \sqrt{(30^2 - 2 \times 6 \times 54)} = \sqrt{252} \approx 15.9$ m/s ≈ 57 km/h. **(This speed calculation is not required.)** A **V2V** warning sent directly from the disabled vehicle (DSRC, or C-V2X in V2X-Direct mode) is latency-sensitive, needs no line of sight and reaches several hundred meters, so the warning arrives well before 81 m. (1 pt for "No" with the numbers; 1 pt for a V2X type with a correct reason.)

Explanation: $v_f^2 = v_0^2 - 2 \cdot a \cdot d$ is normally solved for d with $v_f = 0$. Here the distance is fixed ($d_{remain}$ = 54 m, less than the 75 m needed to stop), so we solve for the final speed: $v_f^2 = 900 - 648 = 252$ $m^2/s^2$, so $v_f = v_{impact} \approx 15.9$ m/s (× 3.6 ≈ 57 km/h). The AV sheds only about 72 % of its 900 $m^2/s^2$ before reaching the obstacle.

**Q3. (10 pts) Classification Metrics (L2)**

Consider a classifier used as a pedestrian detector on 1,000 camera frames, 100 of which contain a pedestrian (positive class). Its confusion matrix is TP = 80, FN = 20, FP = 45, TN = 855.

**(a) (8 pts)** Compute accuracy, precision, recall, F1 score.

**(b) (2 pts)** If the decision threshold is lowered, how do recall and precision typically change? For an AV, is a false negative or a false positive more dangerous here, and why?

ANS:

(a) Accuracy = (TP + TN)/(TP + TN + FP + FN) = (80 + 855)/1000 = **93.5 %**. Precision = TP/(TP + FP) = 80/125 = **0.64**. Recall = TP/(TP + FN) = 80/100 = **0.80**. Note: 93.5 % accuracy looks good although 36 % of the detections are false alarms. F1 = 2PR/(P + R) = 2 × 0.64 × 0.8 / 1.44 ≈ **0.711** (= 2TP/(2TP + FP + FN) = 160/225). (2 pts each)

(b) Lowering the threshold labels more frames as pedestrian: **recall increases** (fewer FN) and **precision typically decreases** (more FP). (1 pt) A **false negative** (a missed pedestrian) is more dangerous because the AV does not brake; a false positive usually causes only unnecessary braking, so safety-critical detectors favor high recall. (1 pt)

**Q4. (20 pts) Convolutional Neural Networks (L3)**
Network: input 32 × 32 × 3 → CONV1 (8 filters, 5 × 5, stride 1, no padding) + ReLU → MAXPOOL (2 × 2, stride 2) → CONV2 (16 filters, 3 × 3, stride 1, padding 1) + ReLU → MAXPOOL (2 × 2, stride 2) → flatten → FC (10 outputs). Each filter and each FC neuron has one bias.

**(a) (6 pts)** Fill in the output shape (H × W × C) and the number of learnable parameters of each layer.

| Layer | Output shape | # parameters |
|---|---|---|
| CONV1 | | |
| MAXPOOL1 | | |
| CONV2 | | |
| MAXPOOL2 | | |
| FC | | |

**(b) (4 pts)** Suppose the CONV1 output (28 × 28 × 8) were instead produced by a fully connected layer from the 32 × 32 × 3 input (ignore biases). How many weights would that need, and how many times more than the 5 × 5 × 3 × 8 weights of CONV1? Name the two properties of convolution that explain the difference.

**(c) (4 pts)** Convolve the 4 × 4 input with the 2 × 2 filter (stride 1, no padding, no flip), then apply ReLU and 2 × 2 max pooling with stride 1. Fill in the tables.

Input:

| | | | |
|---|---|---|---|
| 1 | 2 | 0 | 1 |
| 0 | 1 | 3 | 1 |
| 2 | 1 | 0 | 2 |
| 1 | 0 | 1 | 3 |

Filter:

| | |
|---|---|
| 1 | 0 |
| 0 | −1 |

(i) (2 pts) Convolution output (3 × 3):

| | | |
|---|---|---|
| | | |
| | | |
| | | |

(ii) (2 pts) After ReLU and 2 × 2 max pooling (stride 1) (2 × 2):

| | |
|---|---|
| | |
| | |

**(d) (6 pts)** Replace CONV2 (3 × 3 convolution, input 14 × 14 × 8, output 14 × 14 × 16, ignore biases) by a depthwise separable convolution (3 × 3 depthwise followed by 1 × 1 pointwise).

(i) (3 pts) Number of parameters of the regular and the depthwise separable version.

(ii) (1.5 pts) Number of multiplications for the whole 14 × 14 output of each version.

(iii) (1.5 pts) By what factor is the cost reduced?

ANS:

(a)

| Layer | Output shape | # parameters |
|---|---|---|
| CONV1 | 28 × 28 × 8 | (5·5·3 + 1) × 8 = 608 |
| MAXPOOL1 | 14 × 14 × 8 | 0 |
| CONV2 | 14 × 14 × 16 | (3·3·8 + 1) × 16 = 1,168 |
| MAXPOOL2 | 7 × 7 × 16 | 0 |
| FC | 10 | (7·7·16) × 10 + 10 = 784 × 10 + 10 = 7,850 |

Grading: output shapes 0.6 pt each (3 pts); parameter counts of CONV1, CONV2 and FC 1 pt each (3 pts). Output size = (N − F + 2P)/S + 1: CONV1 (32 − 5)/1 + 1 = 28; CONV2 with padding 1 keeps 14 × 14. The pooling layers have no parameters. Total = 9,626 parameters.

(b) FC: 3,072 inputs × 6,272 outputs = **19,267,584 weights**, about **32,000×** more than CONV1's 600 weights (19,267,584 / 600 ≈ 3.2 × $10^4$). (2 pts) The difference comes from **local connectivity** (each output sees only a 5 × 5 × 3 patch) and **parameter sharing (the same filter is reused at every position). (2 pts)**

(c) Convolution (output = top-left × 1 + bottom-right × (−1) of each 2 × 2 patch):

| | | |
|---|---|---|
| 0 | −1 | −1 |
| −1 | 1 | 1 |
| 2 | 0 | −3 |

After ReLU: [[0, 0, 0], [0, 1, 1], [2, 0, 0]]. After 2 × 2 max pooling with stride 1:

| | |
|---|---|
| 1 | 1 |
| 2 | 1 |

Grading: 2 pts for the 3 × 3 convolution output (−0.5 per wrong entry, minimum 0), 1 pt for ReLU, 1 pt for pooling.

(d) Note: Since almost no one got the depthwise separable convolution correct, I have relaxed grading rules: (i) (3 pts) Number of parameters of the regular convolution. (ii) (3 pts) Number of multiplications of the regular convolution for the whole 14 × 14 output.

Below I show the full correct answer, but only the full convolution parts are counted for grading.

(i) Number of parameters:

**Regular convolution:** each of the 16 filters spans all 8 input channels: 3 × 3 × 8 = 72 weights, so 16 × 72 = **1,152**.

**Depthwise separable: (1) Depthwise:** one 3 × 3 filter per input channel, 8 × (3 × 3) = 72 weights; it filters each channel separately. **(2) Pointwise:** 16 filters of 1 × 1 × 8, 16 × 8 = 128 weights; it mixes the channels. Total = 72 + 128 = **200**. (1.5 pts each)

(ii) Multiplications:

**Regular convolution:** its output is 14 × 14 × 16 = 3,136 elements. Each one is a 3 × 3 × 8 dot product, so 72 MULs, and 3,136 × 72 = 225,792.

**Depthwise separable: Depthwise:** its output is 14 × 14 × 8 = 1,568 elements. Each one is a 3 × 3 × 1 dot product, so 9 MULs, and 1,568 × 9 = 14,112. **Pointwise:** its output is 14 × 14 × 16 = 3,136 elements. Each one is a 1 × 1 × 8 dot product, so 8 MULs, and 3,136 × 8 = 25,088. Total = 14,112 + 25,088 = 39,200. (1.5 pts)

(iii) 1,152 / 200 = **5.76**× fewer parameters and multiplications (= 1/16 + 1/9 ≈ 0.174 of the regular cost). (1.5 pts)

**Q5. (10 pts) Adversarial Attacks (L4.1)**

A toy "image" has 3 pixels, x = (0.5, 0.2, 0.8), with valid pixel values in [0, 1]. The attacker uses an $\ell\infty$ bound $\varepsilon = 0.1$. At $\delta = 0$ the gradient of the loss with respect to the input is $\nabla\delta$ Loss = (2.0, −0.5, 0.0).

**(a) (6 pts) FGSM.**

(i) (3 pts) Compute $\delta$ and the adversarial input $x + \delta$.

(ii) (3 pts) Compute the $\ell_1$, $\ell_2$ and $\ell\infty$ norms of $\delta$.

**(b) (4 pts) Adversarial training.** Fill in the blanks of the adversarial-robust objective "min over _____ of E[ max over _____ of Loss(x + δ, y; θ) ]", and state which gradient is used to update θ (Danskin's theorem).

ANS:

(a) (i) sign(∇) = (+1, −1, 0), so $\delta = \varepsilon \cdot \text{sign}(\nabla)$ = **(0.1, −0.1, 0)** (sign(0) = 0) and x + δ = **(0.6, 0.1, 0.8)**, all within [0, 1]. (2 pts for δ, 1 pt for x + δ)

(ii) $\ell_1 = 0.1 + 0.1 + 0 =$ **0.2**; $\ell_2 = \sqrt{(0.01 + 0.01)} =$ **0.141**; $\ell\infty$ = max = **0.1**. (1 pt each)

(b) min over **θ** of E[ max over **δ ∈ Δ** of Loss(x + δ, y; θ) ]: the inner maximization crafts the worst-case perturbation (attack), the outer minimization trains the classifier. (2 pts) By Danskin's theorem the gradient is taken at the maximizer: ∇θ Loss(x + δ*, y; θ), with δ* = argmax over Δ of Loss(x + δ, y; θ) (valid if the max is solved exactly, so a strong inner attack such as PGD matters). (2 pts)

**Q6. (10 pts) Object Detection: IoU and NMS (L5)**

A detector returns five boxes ($x_1$, $y_1$, $x_2$, $y_2$), in pixels, for two pedestrians in an image. IoU = |A ∩ B| / |A ∪ B|. Greedy NMS: sort the boxes by score, keep the highest-scoring box, suppress every remaining box whose IoU with a kept box is greater than the threshold, and repeat with the highest-scoring remaining box.

| Box | $(x_1, y_1, x_2, y_2)$ | Score |
|---|---|---|
| A | (20, 20, 80, 80) | 0.95 |
| B | (30, 30, 90, 90) | 0.85 |
| C | (22, 22, 82, 82) | 0.70 |
| D | (120, 40, 180, 100) | 0.60 |
| E | (125, 45, 185, 105) | 0.55 |

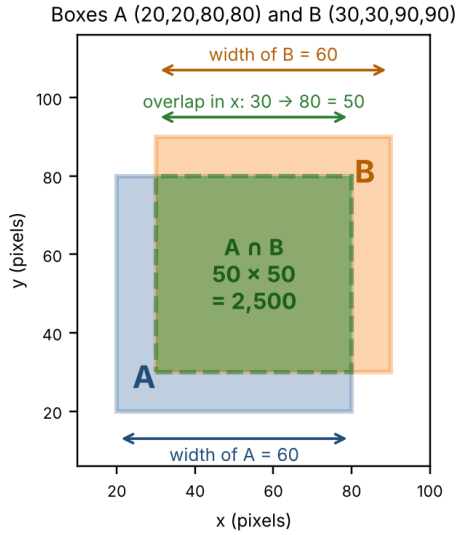
**(a) (4 pts) IoU.** Compute IoU(A, B) and IoU(A, C). Show the intersection and the union areas.

**(b) (4 pts) NMS.** Run greedy NMS with threshold 0.5. For each step, state which box is kept and which boxes are suppressed (you may use IoU(D, E) ≈ 0.72 and IoU(A, D) = 0). Which boxes remain?

**(c) (2 pts) Threshold.** Which boxes remain if the threshold is raised to 0.6, and what is the practical consequence? What goes wrong if the threshold is set too low, e.g. 0.2?

ANS:

(a) Box areas: A = 60 × 60 = 3,600 and B = 60 × 60 = 3,600. The overlap of A and B is x from 30 to 80 and y from 30 to 80, so the intersection = 50 × 50 = 2,500 and the union = 3,600 + 3,600 − 2,500 = 4,700. **IoU(A, B) = 2,500 / 4,700 ≈ 0.53**. (2 pts) For A and C (C is 22…82): intersection = 58 × 58 = 3,364, union = 7,200 − 3,364 = 3,836, **IoU(A, C) = 3,364 / 3,836 ≈ 0.88**. (2 pts) (No need to draw figures below in the exam.)



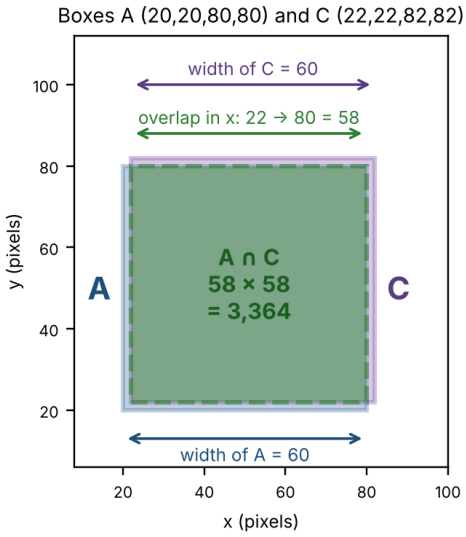

Area(A) = 60 × 60 = 3,600

Area(B) = 60 × 60 = 3,600

Intersection = (80 − 30) × (80 − 30)
= 50 × 50 = 2,500

Union = Area(A) + Area(B) − Intersection
= 3,600 + 3,600 − 2,500 = 4,700

**IoU = Intersection / Union**
= 2,500 / 4,700 ≈ 0.53




Area(A) = 60 × 60 = 3,600

Area(C) = 60 × 60 = 3,600

Intersection = (80 − 22) × (80 − 22)
= 58 × 58 = 3,364

Union = Area(A) + Area(C) − Intersection
= 3,600 + 3,600 − 3,364 = 3,836

**IoU = Intersection / Union**
= 3,364 / 3,836 ≈ 0.88

(b) With threshold 0.5. Step 1: A has the highest score (0.95), so **keep A**. IoU(A, B) = 0.53 > 0.5 and IoU(A, C) = 0.88 > 0.5, so **suppress B and C**; IoU(A, D) = 0 ≤ 0.5, so D and E are not affected by A. (2 pts) Step 2: the highest-scoring remaining box is D (0.60), so **keep D**; IoU(D, E) ≈ 0.72 > 0.5, so **suppress E**. Remaining boxes: **A and D**: one box per pedestrian. (2 pts)

(c) With threshold 0.6. IoU(A, B) = 0.53 ≤ 0.6, so B is no longer suppressed; C (0.88) and E (0.72) are still suppressed. Remaining: **A, B and D**. B is a duplicate detection of the same pedestrian as A (a false positive), because the threshold is too high to remove it. (1 pt) If the threshold is too low (e.g. 0.2), boxes of two different, partly overlapping pedestrians (a crowd) are treated as duplicates, so the second pedestrian is suppressed and missed (a false negative). (1 pt)